

Sumit Raj

Software Engineer

CONTACT

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Portfolio: sumit966-github-io.vercel.app

SKILLS

Languages: Java, Python, JavaScript, C++, SQL
Backend: REST APIs, FastAPI, Node.js, Object-Oriented Programming, Microservices (basic), SDLC
Databases: MySQL, PostgreSQL, MongoDB, SQLite
DevOps and Cloud: Docker, GitHub Actions CI/CD, Git, Linux, GCP
Fundamentals: Data Structures and Algorithms, Problem Solving
Building: Redis caching, rate limiting, unit testing with pytest

EDUCATION

Visvesvaraya National Institute of Technology (VNIT), Nagpur <i>M.Tech, Applied AI and Machine Learning</i>	2024 – May 2026 CGPA: 6.53/10
Dr. D.Y. Patil Institute of Technology, Pune <i>B.E., Computer Science</i>	2019 – 2023 CGPA: 7.89/10

CERTIFICATIONS

- Google Cloud: Cloud Fundamentals and Cloud Security
- Microsoft: Threat Modeling
- HackerRank: 5-Star Java, 4-Star C++, 3-Star Algorithms (300+ problems solved)

EXPERIENCE

Software Engineer Intern, Salesforce (Remote)	Feb 2022 – May 2022
- Built Python bulk data-processing pipelines, cutting manual data handling by approximately 40%. - Integrated Salesforce APIs into optimized backend workflows, improving system efficiency by approximately 25%. - Designed secure data-management modules using object-oriented programming and secure development practices.	

PROJECTS

Scalable URL Shortener Service <i>Python, FastAPI, PostgreSQL, Redis, Docker</i>	2026
- Built a REST API that creates and resolves short links, with PostgreSQL storage and Redis caching for fast redirects. - Added rate limiting, input validation and pytest unit tests. - Containerized with Docker and set up GitHub Actions CI/CD.	
Background Job Queue and Task Scheduler Service <i>Python, FastAPI, PostgreSQL, Docker</i>	2026
- Designed a background job queue with retries, priorities and status tracking exposed through REST APIs. - Applied object-oriented design and unit tests, and containerized the service with Docker.	
RAG-Based Clinical Question Answering System <i>Python, LangChain, ChromaDB, FastAPI, Docker, GitHub Actions</i>	2025
- Built a RAG pipeline for clinical document retrieval with LangChain and ChromaDB, reaching 92% answer relevance on a 100-question benchmark. - Added hybrid search (BM25 + dense embeddings) with Reciprocal Rank Fusion, improving retrieval accuracy by 18%; deployed with FastAPI, Docker and GitHub Actions CI/CD at under 500 ms latency.	
LLM-Powered Security Log Analyzer <i>Python, LangGraph, OpenAI API</i>	2025
- Built a LangGraph agent workflow with the OpenAI API to analyze 50K+ security logs and generate natural-language threat summaries. - Added a self-reflection evaluation loop that kept hallucination under 3% on 200 curated test cases.	
Military Vehicle Detection and Face Authentication Security System <i>Python, YOLOv8, OpenCV, Haar Cascade</i>	Academic Project
- Built a real-time surveillance system with Haar Cascade face authentication at approximately 95% detection accuracy. - Detected and tracked military vehicles with YOLOv8 and OpenCV, sending automated email alerts for security events.	
Brain Tumor Detection from MRI Scans <i>Python, Deep Learning</i>	2024
Trained a deep learning model to classify brain tumors on 3,000+ MRI scans.	
Mobile Botnet Detection System <i>Python, Scikit-learn</i>	2023
Built a machine learning model to detect malicious bot communications with C&C servers by analyzing Android network traffic patterns.	